MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSPBTMU
(INCLUDING VARIATION LSPBTM)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - One copper and one constantan conductor for each pair, uncoated. AWG size 22, class K stranding. The conductor shall have a cross sectional area within plus or minus 4 percent of the nominal circular mils as specified in MIL-C-24643. Temperature electromotive force relationship shall follow ANSI MC96.1. Any necessary joints in the individual strands shall be silver soldered for copper, and welded for constantan.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Ethylene propylene rubber or cross-linked polyethylene insulation. Nominal wall thickness 0.015 inch. Telephone identification code applied by method 3.

Fourth - Each pair shall be one copper and one constantan conductor cabled with a lay not greater than 3 inches.

Fifth - The specified number of pairs (see table I) shall be cabled with a lay not greater than 10 inches in the outer layer. Fillers shall be used in all valleys as necessary to form a round cable. Cabling sequence shall be consecutive, starting with pair no. 1, from center outward.

Sixth - Binder tape.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:56Z
Check the source to verify that this is the current version before use.

Construction variant for type LSPBTMU

Seventh - Cross-linked polyolefin jacket. Cable surface marking required.

Construction variant for type LSPBTM

Seventh - Cross-linked polyolefin jacket.
Eighth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/10	Size 2/	Number of pairs	Overall diameter (inches)				Cold bending diameter maximum (inches)
			Type LSPBTMU		Type LSPBTM		
			min	max	min	max	
-01UN	5	5	0.499	0.540	0.549	0.590	8
-02UN	15	15	.694	.750	.744	.800	10
-03UN	30	30	.907	.980	.957	1.030	13

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSPBTMU and LSPBTM.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
ANSI MC96.1 calibration limits of error.....	± 1°C or ± 0.75 percent
Voltage withstand - volts, root mean square, minimum...	1000
Insulation resistance - megohms/1000 feet, minimum.....	100
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Watertightness - see MIL-C-24643 for limits of water leakage.....	No failure
Group B:	
Cold bending, cable - (see table I for mandrel diameters).....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470).	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum.....	700
Elongation - percent, minimum.....	250

Source: https://assist.dla.mil - Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum.....	75
Elongation - percent of unaged, minimum.....	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum.....	80
Elongation - percent of unaged, minimum.....	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum.....	60
Elongation - percent of unaged, minimum..	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum.....	50
Elongation - percent, minimum.....	50
Shrinkage.....	No failure
Permanence of printing for type LSPBTMU only (jacket) - Cycles, minimum.....	250
Cable filler removability.....	No failure
Heat distortion - percent of unaged, maximum.....	30
Armor - conformance to material construction and coverage (type LSPBTM only).....	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

Aging and compatibility (cable)(125 ± 5°C).....	No failure
Abrasion resistance (jacket/insulation).....	No failure
Acid gas equivalent - percent, maximum	
Jacket.....	2
Fillers.....	2
Insulation.....	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.

QUALIFICATION INSPECTION: (Continued)

	Requirements
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	---
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
5	2500
15	1500
30	1000

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0927)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T23:56Z
Check the source to verify that this is the current version before use.